

Heme B Biosynthesis Through Alternative Entry and Oxidation Routes: A Cross-Taxon Review of the Protoporphyrin-Dependent System

A review-style synthesis for a molecular-biology audience. Search date: 2026-07-21. Primary anchor references are Dailey et al. 2017

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[P 28123057](#)
and Dailey et al. 2015
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1. Executive summary

Heme B (protoheme IX, Fe(II)–protoporphyrin IX) is an ancient, near-universal cofactor. This review treats one coherent biological system: the **protoporphyrin-dependent (PPD)** route to heme B, in which a shared tetrapyrrole "trunk" builds coproporphyrinogen III, this is converted through **protoporphyrinogen IX** to **protoporphyrin IX**, and a single, mechanistically conserved ferrochelatase inserts ferrous iron to make heme B. The system is best understood as a *reusable modular scaffold*: two mutually exclusive **entry chemistries** to 5-aminolevulinate (ALA) — the C5 glutamyl-tRNA route (GluTR + GSA-aminomutase) and the C4/Shemin route (ALA synthase) — feed an invariant six-enzyme trunk, after which two **redox branch points** (coproporphyrinogen oxidation; protoporphyrinogen oxidation) are each solved by *non-orthologous, convergently evolved* enzyme families selected largely by oxygen availability. All valid combinations converge on protoporphyrin IX and then on a species-neutral ferrochelatase

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Three points dominate the modern picture. **(i)** The PPD route is only one of three natural terminal architectures; the **siroheme-dependent (SHD)** route of Archaea/sulfate reducers and the **coproporphyrin-dependent (CPD)** route of Gram-positive bacteria are separate systems that must not be conflated with it

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(ii) At the two late redox steps, evolution has repeatedly *replaced the protein without changing the reaction* — HemF vs HemN at coproporphyrinogen oxidation, and HemG vs HemJ vs HemY at protoporphyrinogen oxidation — the clearest examples of non-orthologous gene displacement in cofactor metabolism

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(iii) The endpoints are deeply conserved (the uroporphyrinogen III core and ferrochelatase), whereas the entry and late steps are the parts that vary, are lost, or are laterally transferred across lineages.

2. Definition and biological boundaries

What is inside the system. The PPD system comprises every enzymatic step required to build heme B *via protoporphyrin IX*:

- **Entry (route-specific ALA formation):** either C5 (glutamyl-tRNA reductase, HemA/GluTR → glutamate-1-semialdehyde 2,1-aminomutase, HemL/GSA-AM) or C4/Shemin (5-aminolevulinate synthase, ALAS/HemA-of- α -proteobacteria).
- **Shared trunk (ALA → coproporphyrinogen III):** porphobilinogen synthase (ALA dehydratase, HemB); hydroxymethylbilane synthase (porphobilinogen deaminase, HemC); uroporphyrinogen-III synthase (HemD); uroporphyrinogen decarboxylase (HemE).
- **Late redox module:** coproporphyrinogen III oxidase — O₂-dependent HemF **or** O₂-independent HemN; then protoporphyrinogen IX oxidase — HemG **or** HemJ **or** HemY.
- **Convergence/terminal step:** protoporphyrin ferrochelatase (HemH), Fe(II) insertion → heme B.

What is outside the system (near-neighbors and common confusions).

- **The coproporphyrin-dependent (CPD) pathway** of Actinobacteria and Firmicutes. These Gram-positive lineages **cannot make protoporphyrin**; they oxidize coproporphyrinogen to

coproporphyrin (CgoX, confusingly also called "HemY"), insert iron to make coproheme, and *then* decarboxylate coproheme to heme B via coproheme decarboxylase (ChdC/HemQ)

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Because it inverts the order of ferrochelation and side-chain decarboxylation and never forms protoporphyrin IX, CPD is a **separate system** that shares only the early trunk. The shared "HemY" name and shared trunk enzymes make this the single most important boundary to police.

- **The siroheme-dependent (SHD) / "alternative" pathway** of Archaea and many sulfate-reducing bacteria, which routes uroporphyrinogen III → precorrin-2 → siroheme → (Fe-coproporphyrin/didecarboxysiroheme) → heme B by oxygen-independent Ahb/Nir enzymes, bypassing both coproporphyrinogen oxidase and PPOX entirely

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This is the deepest branch and is outside PPD scope.

- **Downstream and sibling tetrapyrrole branches:** heme *o*/heme *a* (heme O synthase, heme A synthase), siroheme, chlorophyll/bacteriochlorophyll (Mg-chelatase branch at protoporphyrin IX), cobalamin, coenzyme F430, and phycobilins. Protoporphyrin IX is the *last* intermediate shared with the chlorophyll branch (via Mg-chelatase), so the chlorophyll branch shares the trunk but diverges before ferrochelatase.
- **Heme trafficking, degradation, and regulation:** heme oxygenases (HMOX1/2), heme transporters and chaperones, heme-regulated transcription (e.g., Irr, HemA proteolysis), and disease states (the human porphyrias) are physiologically intertwined but are *not* biosynthetic steps of the PPD system and are treated as adjacent processes here.

Competing definitions in the literature. Two definitional tensions recur. First, older reviews assumed a *single* universal heme pathway; this is now known to be wrong — three pathways exist

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so any "heme biosynthesis" claim must specify the branch. Second, enzyme names are overloaded across branches (**HemY** = PPOX in PPD but = coproporphyrinogen oxidase CgoX in CPD; **HemA** = GluTR in most bacteria but historically = ALAS in α -proteobacteria; **HemH** = protoporphyrin ferrochelatase in PPD but coproporphyrin ferrochelatase CpfC in CPD). Mixing organism data under one gene label is a frequent error.

3. Mechanistic overview

Figure 1 summarizes the modular architecture: two alternative ALA-entry routes feed one obligatory trunk, which passes through two oxygen-selected redox branch points before all realizations converge at protoporphyrin IX and ferrochelatase; the coproporphyrin- and siroheme-dependent neighbors are shown as out-of-scope.

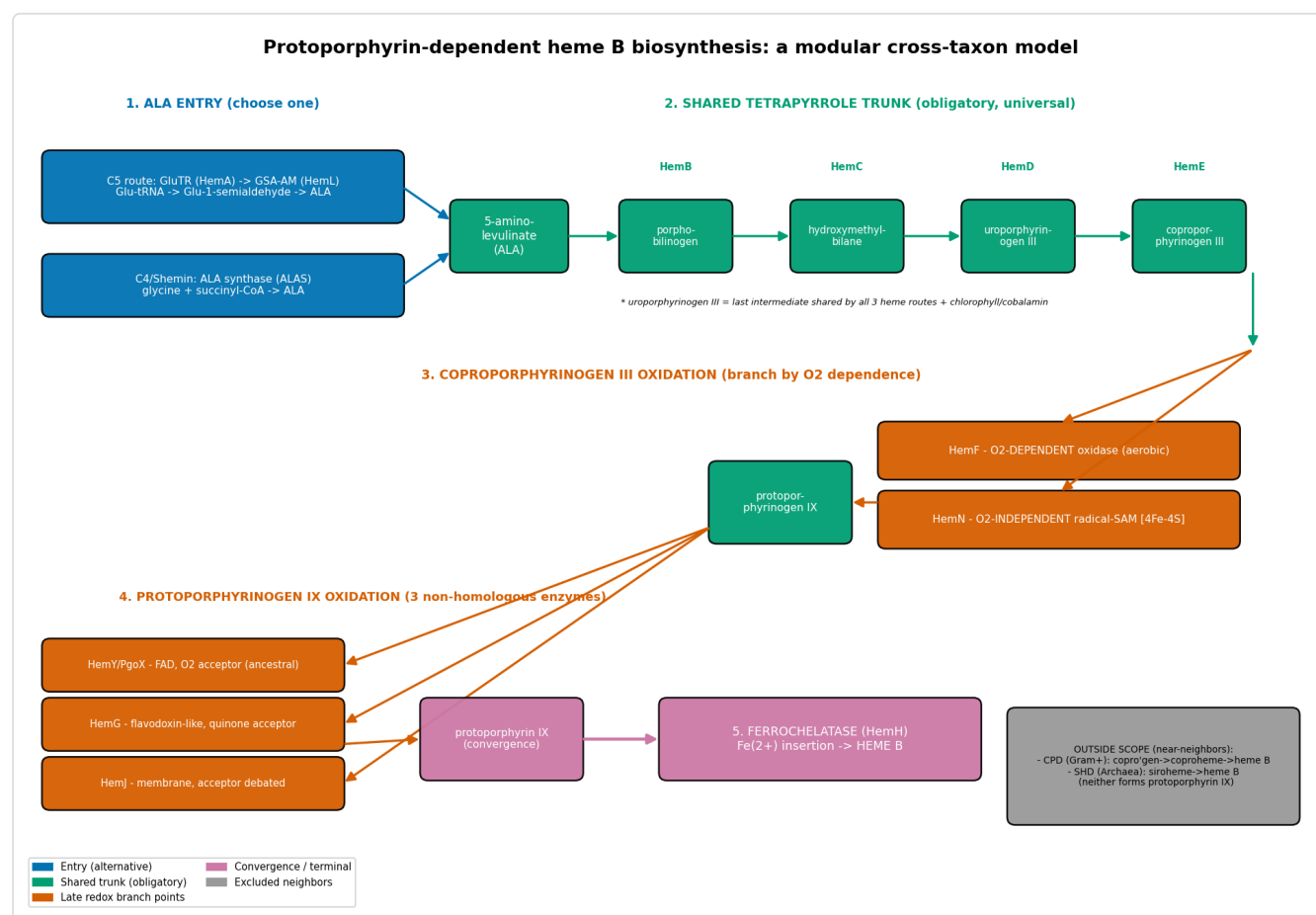


Figure 1. Modular architecture of protoporphyrin-dependent heme B biosynthesis. Two mutually exclusive entry routes (C5 glutamyl-tRNA via GluTR/HemL; C4/Shemin via ALAS) converge at 5-aminolevulinic acid and feed a conserved six-enzyme trunk (HemB-HemE) to coproporphyrinogen III. Two oxygen-selected redox branch points - coproporphyrinogen oxidation (O₂-dependent HemF vs radical-SAM HemN) and protoporphyrinogen oxidation (the non-homologous HemG, HemJ, and HemY) - converge on protoporphyrin IX, which a species-neutral ferrochelatase (HemH) metalates to heme B. The coproporphyrin-dependent (Gram-positive) and siroheme-dependent (archaeal) routes are shown as excluded near-neighbors that reach heme B without forming protoporphyrin IX.

Best current model (sequence of events). The obligatory backbone, in order, is:

ALA → (×2) porphobilinogen → (×4, linear tetrapyrrole) hydroxymethylbilane → uroporphyrinogen III → coproporphyrinogen III → protoporphyrinogen IX → protoporphyrin IX → heme B.

1. **ALA formation (conditional, route-specific).** *Either* ALAS condenses glycine + succinyl-CoA (a PLP-dependent, decarboxylative Claisen-type condensation; product release is rate-limiting;

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or GluTR reduces glutamyl-tRNA(Glu) to glutamate-1-semialdehyde (NADPH-dependent) and GSA-aminomutase (also PLP-dependent) isomerizes it to ALA

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These are alternatives; an organism runs one.

2. **ALA condensation (obligatory).** Porphobilinogen synthase (HemB) asymmetrically condenses two ALA molecules to the monopyrrole porphobilinogen. It is a classic morphoein — its oligomeric equilibrium and metal usage vary phylogenetically: mammalian/*E. coli* enzymes are Zn-dependent octamers (with a stimulatory Mg in *E. coli*/plants), whereas some enzymes are metal-independent hexamers

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3. **Hydroxymethylbilane formation (obligatory).** Hydroxymethylbilane synthase / porphobilinogen deaminase (HemC) uses a unique dipyrromethane cofactor to processively polymerize four porphobilinogens into the linear tetrapyrrole hydroxymethylbilane.
4. **Ring closure (obligatory).** Uroporphyrinogen-III synthase (HemD) inverts ring D and cyclizes to the physiological, asymmetric uroporphyrinogen III. (Non-enzymatic closure yields the "wrong" uroporphyrinogen I — a spontaneous side reaction the cell must suppress.) Uroporphyrinogen III is the **last intermediate shared by all three heme pathways** and by the corrin/siroheme branches.
5. **Peripheral decarboxylation (obligatory).** Uroporphyrinogen decarboxylase (HemE) sequentially removes four acetate carboxyls → coproporphyrinogen III. This is the last step of the universal bacterial core before the PPD/CPD split.

6. **Coproporphyrinogen oxidation (obligatory step, alternative enzymes).** Two propionates on rings A and B are oxidatively decarboxylated to vinyls → protoporphyrinogen IX. Aerobically by **HemF** (O₂-dependent); anaerobically by **HemN**, a [4Fe-4S] radical-SAM enzyme that generates a 5'-deoxyadenosyl radical to abstract hydrogen and drive decarboxylation without O₂

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7. **Protoporphyrinogen oxidation (obligatory step, alternative enzymes).** Six-electron aromatization of the macrocycle → protoporphyrin IX, by one of three unrelated enzymes: **HemY** (FAD, O₂ as acceptor), **HemG** (flavodoxin-like, quinone as acceptor, anaerobic-competent), or **HemJ** (membrane protein, mechanism least understood)

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8. **Ferrochelation (obligatory, convergence).** Ferrochelatase (HemH) distorts (saddles) protoporphyrin IX, deprotonates the pyrrole nitrogens, inserts Fe(II), and releases a flatter metalated product; metal specificity is set at the product-release step through a conserved π -helix switch

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Obligatory vs conditional vs accessory. Steps 2–8 (the trunk + late module + ferrochelatase) are **obligatory** for PPD heme B. The *choice* of enzyme at steps 1, 6, and 7 is **conditional** (set by lineage and by oxygen tension). Genuinely **accessory** elements include regulatory appendages (e.g., the ALAS C-terminal extension; the FLU/GluTR feedback module in plants) and lineage-specific fusion or targeting domains, none of which change the chemistry.

4. Major molecular players and active assemblies

Step	Enzyme (gene)	Family / cofactor	Notable structural/mechanistic feature	Key refs (PMID)
ALA (C4)	ALA synthase (ALAS/hemA in α -proteobacteria)	PLP, homodimer	Glycine + succinyl-CoA condensation; active-site loop gates turnover; product release rate-limiting	31345668, 26605136
ALA (C5)	Glutamyl-tRNA reductase (hemA/GluTR)	NADPH; binds Glu-tRNA	V-shaped dimer; substrate channeling to HemL; feedback target (heme, FLU)	40315358, 30863751
ALA (C5)	Glutamate-1-semialdehyde 2,1-aminomutase (hemL)	PLP aminomutase	Intramolecular amino transfer to ALA; gabaculine-sensitive (diagnostic of C5 route); inhibitor-resistant point mutants map to hemL	28123057, 9267435, 10350057
Trunk 1	Porphobilinogen synthase (hemB/ALAD)	Zn and/or Mg; morpheein	Octamer/hexamer equilibrium; variable metal usage	7819203, 15555082, 21383008
Trunk 2	Hydroxymethylbilane synthase (hemC/PBGD)	Dipyrromethane cofactor	Processive tetramerization of PBG	28123057
Trunk 3	Uroporphyrinogen-III synthase (hemD)	—	Ring-D inversion/cyclization to III isomer	28123057
Trunk 4	Uroporphyrinogen decarboxylase (hemE)	—	Four sequential decarboxylations	28123057
CPO	O ₂ -dependent coproporphyrinogen oxidase (hemF)	O ₂ as oxidant	Aerobic vinyl formation	28123057
CPO	O ₂ -independent coproporphyrinogen dehydrogenase (hemN)	Radical SAM, [4Fe-4S], 2× SAM, NAD(P)H	First radical-SAM crystal structure;	12114526, 14633981, 16218869

Step	Enzyme (gene)	Family / cofactor	Notable structural/mechanistic feature	Key refs (PMID)
			CXXXCXXC cluster motif	
PPOX	HemY / PgoX (PPOX)	FAD; O ₂ acceptor	Ancestral, ubiquitous, only eukaryotic PPOX; acifluorfen-sensitive	25108393, 20823222
PPOX	HemG	Flavodoxin-like; quinone acceptor	γ -Proteobacterial; couples to respiratory quinone pool	25108393
PPOX	HemJ	Integral-membrane; heme <i>b/b</i> -type	α -Proteobacterial/cyanobacterial; mosaic distribution	25108393, 29925590, 20823222
Terminal	Protoporphyrin ferrochelatase (hemH)	Some carry [2Fe-2S]	Porphyrin saddling; π -helix product-release switch controls metal specificity	21776189, 19703464, 29481781

Assemblies and channeling. Several steps are physically coupled. GluTR and GSA-aminomutase form a complex/channel that hands off the reactive glutamate-1-semialdehyde (

[P 30863751](#)

In many organisms the late, membrane-associated steps (HemN/HemG/HemJ and ferrochelatase) are localized near the membrane and, in α -proteobacteria/mitochondria, spatially organized to hand protoporphyrinogen/protoporphyrin/iron along the inner membrane. HemG and HemJ tie the aromatization step directly to the respiratory chain by using quinone or membrane redox as the electron sink.

5. Evolutionary and cell-biological variation

Across lineages. - **Entry chemistry:** C5 (GluTR/GSA-AM) is the broad ancestral bacterial/archaeal/plant route; C4/ALAS is restricted to **α -proteobacteria, fungi, and animals** and is generally regarded as the derived route (

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Because mitochondria descend from α -proteobacteria, animals/fungi inherited ALAS; plants retained the cyanobacteria-derived C5 route in the plastid. - **Late redox module:** HemF (aerobes) vs HemN (facultative/anaerobes); many facultative organisms encode **both** and switch by oxygen tension. At PPOX, HemG is largely confined to γ -Proteobacteria, HemJ appears to have originated in α -Proteobacteria and spread to Cyanobacteria, and HemY is ubiquitous and the sole eukaryotic form (P 25108393).

Cyanobacteria are a natural experiment: all three PPOX types occur across the phylum in a mosaic, occasionally two in one genome, with cross-complementation demonstrated experimentally (P 36357749).

- **Losses:** Many host-associated bacteria and some eukaryotes (e.g., certain parasites, and organisms auxotrophic for heme) have lost part or all of the pathway and import heme; malaria parasites present a chimeric "nested" heme economy combining parasite synthesis, truncated host-erythrocyte capacity, and host-hemoglobin-derived heme (P 42198604).

Compartments, cell types, and states (eukaryotes). In animals the pathway is split between **mitochondrion** (ALAS, and the terminal HemN-absent module: coproporphyrinogen oxidase, PPOX-HemY, ferrochelatase — all mitochondrial/inner-membrane) and **cytosol** (HemB–HemE), requiring transport of intermediates across the mitochondrial membranes. In plants the entire pathway is largely **plastidic**. Tissue/stage specialization is pronounced: erythroid cells use the dedicated **ALAS2** isoform under iron/erythroid control, whereas housekeeping **ALAS1** serves other tissues; gain-of-function ALAS2 causes X-linked protoporphyria and loss-of-function causes X-linked sideroblastic anemia (P 38888931).

In legume nodules, plant GluTRs (notably GluTR2) dominate heme supply for leghemoglobin, coordinating host and symbiont pathways (P 40315358).

These distinctions caution against extrapolating one cell type's regulation to all biology.

Alternative routes to the same outcome. Within PPD, the O₂-dependence branches at CPO and PPOX are alternative molecular means to identical chemical outcomes. Zooming out, CPD and SHD are wholly alternative *routes to heme B itself*, underscoring that "make heme B" has been solved at least three times.

6. Conservation, origin, and best ancestral representatives

Deepest origin. Tetrapyrrole biosynthesis is ancient, predating the divergence of the domains; the shared core to **uroporphyrinogen III** is common to heme, chlorophyll, siroheme, cobalamin, and F430 synthesis and is therefore among the oldest conserved biosynthetic modules

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The **SHD/siroheme route is argued to be the most ancient route to heme**, consistent with an anaerobic origin, with the bacterial coproporphyrinogen-forming step (HemE) added later, and the PPD "protoporphyrin" terminus emerging with (or adapting to) oxygenic conditions

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Ancient/conserved vs derived parts. - *Ancient and conserved:* the trunk (HemB–HemE) and **ferrochelatase**, which share deep homology and mechanism across the tree; the C5 entry route. - *Later elaborations/replacements:* the C4/ALAS entry (α -proteobacterial/eukaryotic); the O₂-dependent enzymes HemF and HemY as adaptations to aerobic environments; HemG and HemJ as lineage-specific **non-orthologous replacements** of PPOX.

Best representatives for the ancestral role. For PPOX, **HemY is the best proxy for the ancestral function** (ubiquitous; sole eukaryotic form)

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For the O₂-independent chemistry that likely predates the Great Oxidation Event, **HemN** (radical SAM) exemplifies the ancestral anaerobic solution at CPO; radical-SAM chemistry is itself an ancient enzyme superfamily

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For ferrochelatase, bacterial and human enzymes are structurally congruent, so either serves

to understand the conserved distortion/ π -helix mechanism

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7. Constraints, dependencies, and failure modes

Ordering constraints. The backbone is a strict linear dependency chain: each intermediate is the substrate of the next (ALA→PBG→HMB→uro'gen III→copro'gen III→proto'gen IX→proto IX→heme B), so steps cannot be reordered. Ring closure (HemD) must precede decarboxylation, and coproporphyrinogen oxidation must precede protoporphyrinogen oxidation — the substrate identities enforce order.

Mutually exclusive / conditional choices. - **Entry:** an organism runs C4 *or* C5, not both (the choice is genomically fixed). - **Oxygen dependence:** HemF/HemY require O₂; HemN/HemG operate anaerobically. In strict anaerobes the O₂-dependent enzymes are non-viable options; in facultative organisms the redundant pair is regulated by oxygen. This is the strongest evidence *ruling out* otherwise plausible paths: an obligate anaerobe cannot use HemF or O₂-coupled HemY, so those combinations are excluded on thermodynamic/mechanistic grounds. - **Compartmentalization (eukaryotes):** the split between cytosolic and mitochondrial/plastidic steps forces transport dependencies; failure of an intermediate transporter phenocopies enzyme loss. - **Branch exclusivity at protoporphyrin IX:** protoporphyrin IX is partitioned between ferrochelatase (Fe → heme) and Mg-chelatase (Mg → chlorophyll) in phototrophs; misallocation is a regulated, competitive node.

Failure modes. Blockages cause accumulation of the upstream intermediate. In humans, each enzyme deficiency defines a **porphyria** with a characteristic biochemical signature (e.g., PBGD deficiency → acute intermittent porphyria; PPOX deficiency → variegate porphyria; coproporphyrinogen oxidase deficiency → hereditary coproporphyria; ferrochelatase deficiency → erythropoietic protoporphyria)

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Accumulated porphyrinogens auto-oxidize to phototoxic porphyrins generating reactive oxygen species

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([P 42197273](#)).
 The reactivity of protoporphyrin IX itself — heme's "dark twin" — makes flux completion, not merely pathway presence, the key constraint
 ([P 42197273](#)).
 This chemistry is exploited therapeutically: exogenous ALA drives PpIX accumulation for photodynamic therapy and fluorescence-guided surgery
 ([P 41070141](#)).
 ([P 38852796](#)).

8. Controversies and open questions

- **HemJ mechanism and cofactor.** HemJ is the least understood PPOX; its precise redox mechanism, cofactor/heme content, and terminal electron acceptor remain incompletely defined despite functional confirmation in cyanobacteria

([P 29925590](#)).
 ([P 20823222](#)).
 Whether it is truly O₂-dependent, quinone-coupled, or flexible is unresolved.

- **The [2Fe-2S] cluster of ferrochelatases.** Present in human and some bacterial ferrochelatases, its role (catalytic, structural, redox/iron-sensing) is still debated

([P 42184855](#)).
 ([P 34402499](#)).

- **Iron delivery to ferrochelatase.** How Fe(II) is channeled to the buried active site (a putative iron channel, chaperone hand-off, or membrane-coupled delivery) is not settled

([P 22897320](#)).
 ([P 34402499](#)).

- **Evolutionary polarity and lateral transfer.** The intricate, partly conflicting phylogenies of HemG/HemJ/HemY (mosaic distributions, apparent horizontal transfer) complicate simple ancestral reconstructions

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The rooting of the three heme pathways (SHD-first model) rests substantially on genome distribution/parsimony rather than direct ancestral-state experiments

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- **Cross-organism generalization.** Much mechanistic detail comes from a few models (*E. coli* HemN; human/yeast ferrochelatase and ALAS2; cyanobacterial PPOX). PBGS metal usage and oligomeric state vary enough between organisms

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that assay conditions and organism choice materially affect conclusions — a caution against universal claims.

- **Channeling and metabolon organization.** The degree to which late membrane steps form a physical metabolon (and whether intermediates are ever free) is largely inferential.

Most important open questions: (1) a definitive HemJ mechanism; (2) the physiological role of the ferrochelatase Fe–S cluster and the iron-delivery route; (3) experimentally grounded ancestral reconstruction of the entry and late-step modules; (4) how flux is partitioned at protoporphyrin IX between Fe- and Mg-chelation in phototrophs under stress.

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Scope caveat: This review is deliberately restricted to the protoporphyrin-dependent route to heme B. Statements about mechanism are anchored where possible in the model organism from which they derive; the coproporphyrin-dependent and siroheme-dependent routes, the chlorophyll/siroheme/cobalamin sibling branches, and heme trafficking/degradation are treated as adjacent systems and are not covered in mechanistic depth.